

James Williams

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Education

University of Central Florida – GPA: 3.92

Expected Graduation – Spring 2027

Bachelor of Science in Computer Science

Orlando, FL

Relevant Coursework: Data Structures & Algorithms, OOP, Database Design, Calculus I, II, III, Linear Algebra

Organizations & Clubs: ColorStack, NSBE, KnightHacks, Events & Outreach Coordinator @ F1 AI Racer Research

Technical Skills

Languages: Python, C/C++, Java, MySQL, PostgreSQL, CSS, HTML, JavaScript, TypeScript, Linux

Frameworks/Libraries: React, Node.js, Flask, Tailwind CSS, Matplotlib, NumPy, Pandas, FastAPI, REST API

Developer Tools: Git, GitHub, VS Code, Docker, Firebase, PyCharm, Azure, MongoDB, CI/CD, AI/ML

Experience

Brown and Brown Insurance

May 2025 – Aug 2025

Software Engineering Intern

Daytona Beach, FL

- Architected and tested an enterprise-level global dashboard using **Python**, **React**, **Leaflet.js**, and **PostgreSQL** to optimize infrastructure visibility across more than **700** international corporate office locations
- Enhanced outage response time by **40%** through the implementation of real-time geospatial tracking features, live system health monitoring, and map-based alert visualization tools to help operators respond more efficiently
- Collaborated in **Agile** sprints and contributed to a scalable backend pipeline for real-time global power tracking integrating **PostgreSQL**, **REST APIs**, and distributed data ingestion workflows across multiple regions

Gary S. May Shadow Initiative Program

Oct 2024 – April 2025

Front-End Developer Intern

Orlando, FL

- Integrated and launched an internal web portal using **React**, **Tailwind**, and **REST APIs** to centralize event management, data sharing, announcements, and resource distribution for over **50** regional chapter leaders
- Streamlined reporting and event tracking by integrating the **Google Sheets API** into internal workflows, improving data accessibility across multiple teams and reducing manual update tasks by over **60%**
- Spearheaded technical solutions to improve digital outreach, chapter coordination, and regional tool deployment, resulting in a **25%** year-on-year increase in engagement across the entire Southeast NSBE network

Undergraduate Research Study

Sep 2023 – April 2024

Software Developer Intern

Davie, FL

- Analyzed statewide Florida public school performance datasets using **Python**, **pandas**, and **NumPy** to evaluate correlations between school start times and test score outcomes across over **500** institutions
- Performed regression analysis, statistical hypothesis testing, and extensive data cleaning in **Python** and **Excel**; finalized findings were submitted to the Florida Undergraduate Research Journal for peer review
- Automated a scalable preprocessing pipeline for over **10,000** rows of raw test score data, reducing analysis time by **35%** enabling rapid iteration on regression models, visualizations, and statistical reporting

Projects

InvestEd | *Next.js, TypeScript, Zustand, shadCN, Gemini 2.5 Flash, Vercel, Tailwind*

Oct 2025

- Placed **3rd** out of 28 teams in the Knight Hacks 2025 OneEthos AI for Financial Empowerment Challenge
- Engineered an AI chatbot with **Gemini 2.5 Flash API** to generate personalized investing advice
- Built a Smart HYSA Finder with APY rankings and an Investment Dashboard with asset performance analytics

MOVE | *Next.js, TypeScript, Tailwind, WebGPU/WebGL, MediaPipe, TensorFlow, Computer Vision*

Sep 2025

- Built a real-time computer vision game using **MediaPipe Pose** to track movement at 60 FPS with <80ms latency
- Developed game logic in **Next.js** to handle collisions, scoring, and a pausing feature when players leave the frame
- Implemented persistent high-score storage and responsive UI, tested by **40+** users during ShellHacks 2025

InsureFair | *React, Node.js, JavaScript, JSX, Tailwind CSS*

Sep 2024

- Launched a **React** app at ShellHacks to estimate insurance overpayment risk using life expectancy and premiums
- Worked with a 4-person team to design a responsive UI with real-time financial computations
- Improved insurance risk clarity by **30%** through streamlined user interaction and visual output